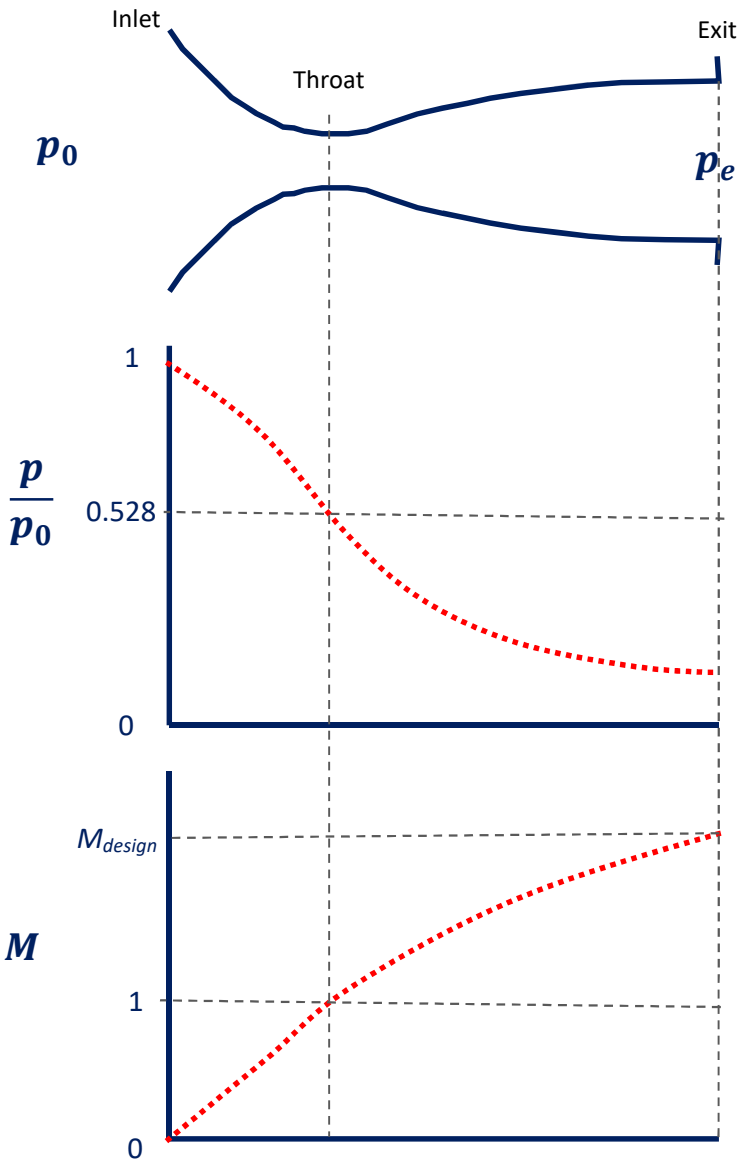
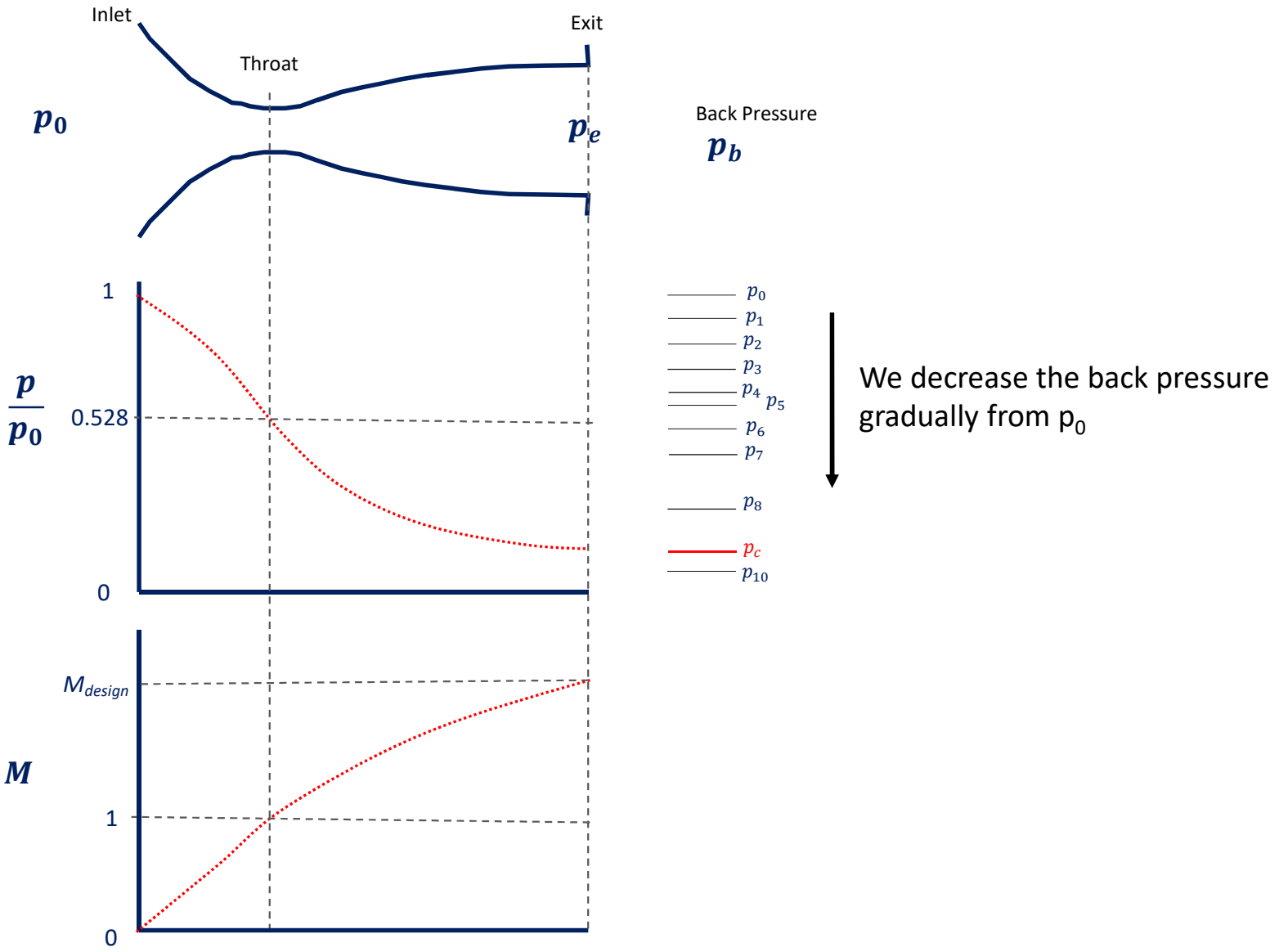
Isentropic Supersonic Nozzle Flow

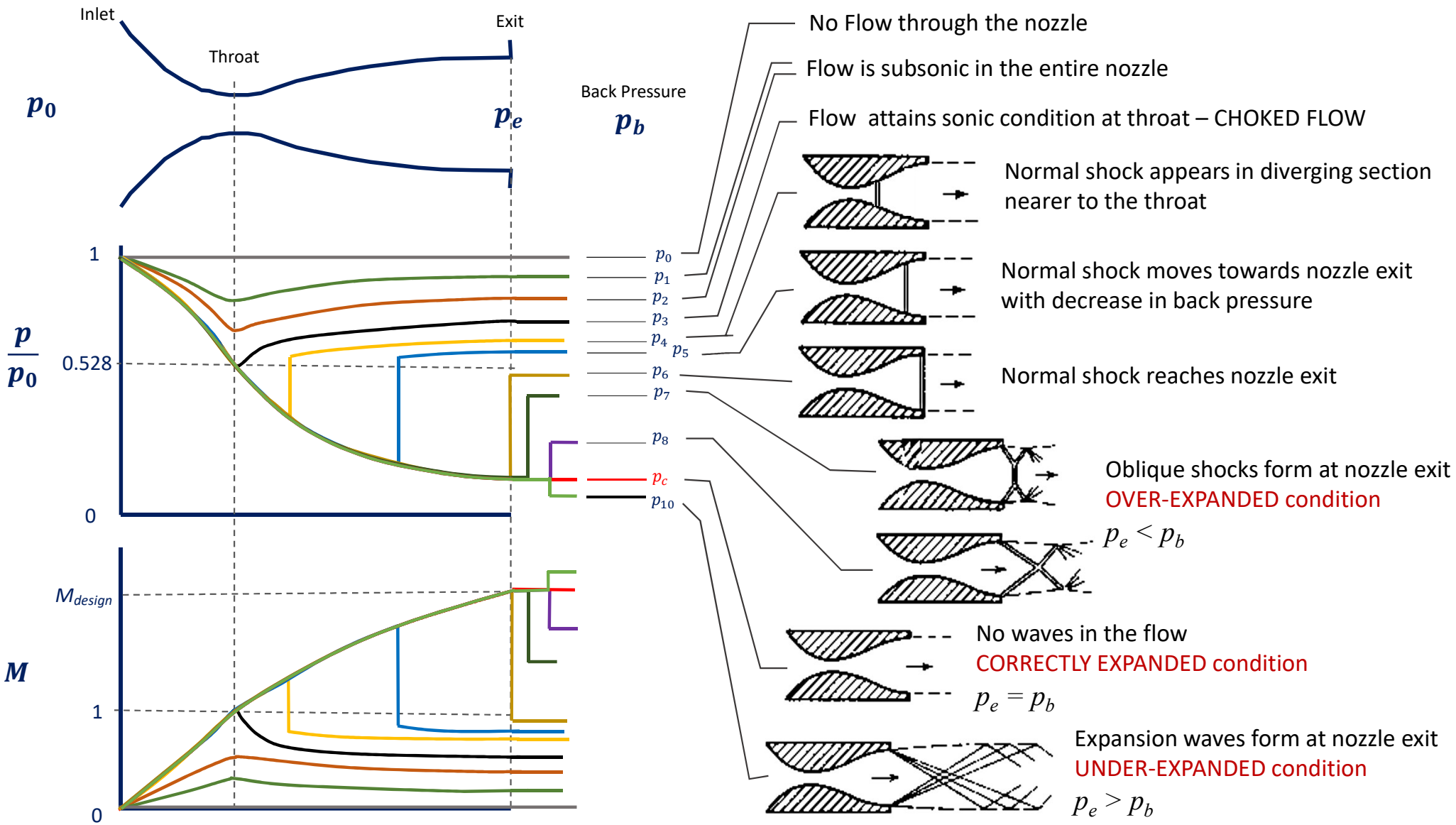


Correctly Expanded Nozzle condition

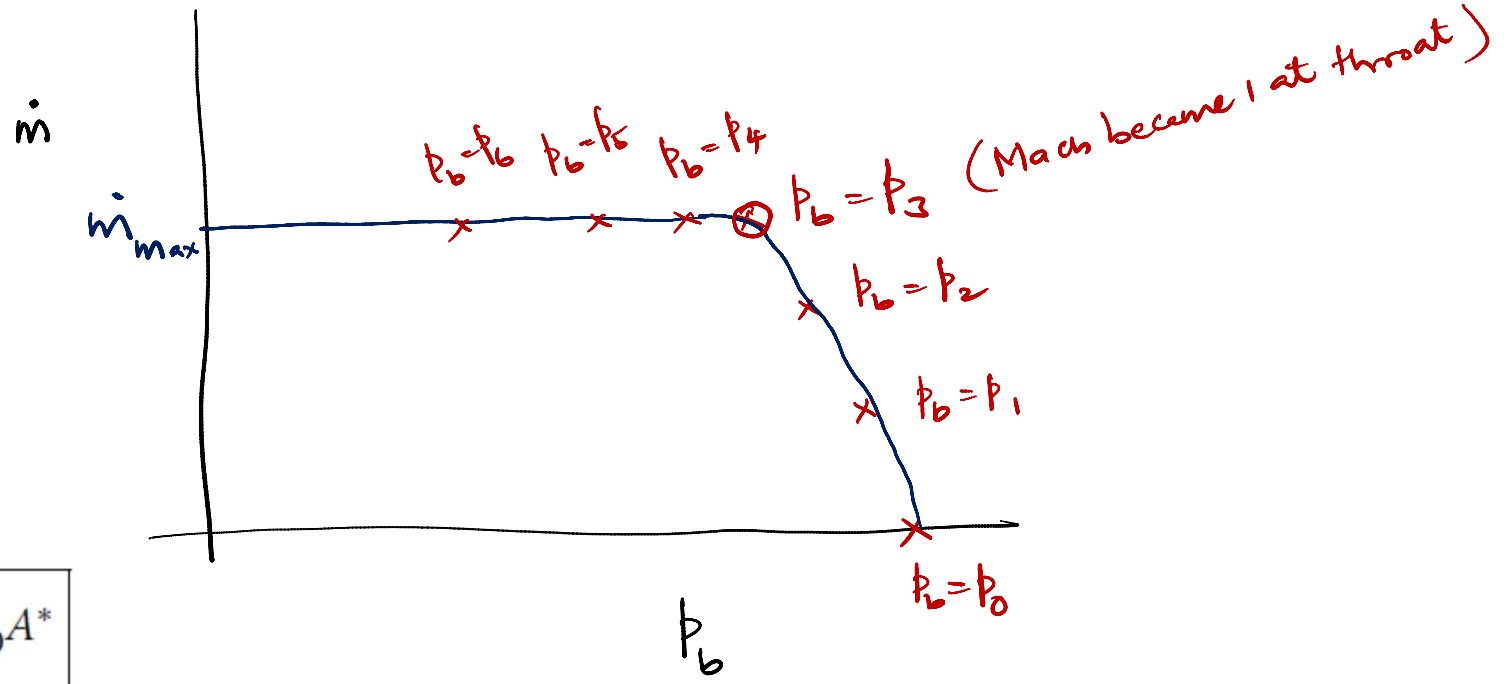
For example,
For generating Supersonic flow of Mach 2 from the nozzle,
$A_e/A^* = 1.69$ and
$p_e = p_b = 0.128p_0$
(From isentropic flow tables)

The Effect of Different Pressure Ratios Across a Given Nozzle


Inlet
Throat
Exit
p_0
p_e
Back Pressure
p_b
1
$\frac{p}{p_0}$
0.528
0
p_0
p_1
p_2
p_3
p_4
p_5
p_6
p_7
p_8
p_c
p_{10}
We decrease the back pressure gradually from p_0
M_{design}
M
1
0


Inlet
Throat
Exit
p_0
p_e
Back Pressure
p_b
$\frac{p}{p_0}$
1
0.528
0
M
M_{design}
1
0
p_0
p_1
p_2
p_3
p_4
p_5
p_6
p_7
p_8
p_c
p_{10}
No Flow through the nozzle
Flow is subsonic in the entire nozzle
Flow attains sonic condition at throat – CHOKED FLOW
Normal shock appears in diverging section nearer to the throat
Normal shock moves towards nozzle exit with decrease in back pressure
Normal shock reaches nozzle exit
Oblique shocks form at nozzle exit
OVER-EXPANDED condition
$p_e < p_b$
No waves in the flow
CORRECTLY EXPANDED condition
$p_e = p_b$
Expansion waves form at nozzle exit
UNDER-EXPANDED condition
$p_e > p_b$

Mass Flow Rate through CD Nozzle – Choked Condition



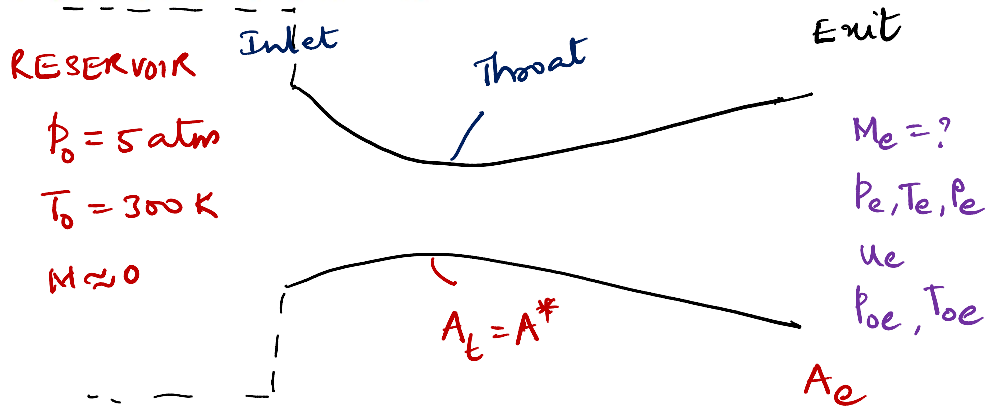
$$\dot{m}_{\max} = \boxed{\frac{0.6847}{\sqrt{RT_0}} p_0 A^*}$$

* Maximum mass flow rate occurs when Mach number becomes 1 at the throat ⟶ CHOKED CONDITION

- When the back pressure is equal to the inlet pressure p_0, there is no flow in the nozzle.
- As the back pressure is reduced from p_0, the pressure at the throat decreases and the mass flow rate increases gradually.
- At a certain value of back pressure, the throat pressure becomes $p^*(=0.528p_0)$ and the throat Mach number becomes 1. The mass flow rate in the nozzle attains m_{max}. – CHOKED CONDITION
- With a further decrease in the back pressure, the throat pressure, throat Mach number and mass flow rate of the nozzle does not vary since the nozzle has attained choked condition.

Problems on nozzle flows

Problem: The reservoir pressure and temperature for a convergent-divergent nozzle are 5 atm and 300 K respectively. The flow is expanded isentropically to supersonic speed at the nozzle exit. If the exit-to-throat area ratio is 2.193, calculate the following properties at the exit: M_e, p_e, T_e, ρ_e, u_e, $p_{0,e}$, $T_{0,e}$.



$$\frac{A_e}{A^*} = 2.193$$

$$\frac{A_e}{A^*} = 2.193 \xrightarrow{\text{Isentropic tables}} M_e = 2.3$$

- $\frac{p_e}{p_{oe}} = 0.08$
- $\frac{T_e}{T_{oe}} = 0.486$

Since the flow is isentropic, p_0 and T_0 are constant in the entire flow

$\Rightarrow p_{oe} = p_0$ and $T_{oe} = T_0$

$$p_e = p_{oe} \times 0.08 = 5 \times 0.08 = 0.4 \text{ atm}$$

$$T_e = T_{oe} \times 0.486 = 300 \times 0.486 = 145.8 \text{ K}$$

$$p_e = \rho_e R T_e \Rightarrow \rho_e = \frac{p_e}{RT_e} = \frac{0.4 \times 101325}{287 \times 145.8} =$$

$$\Rightarrow \rho_e = 0.97 \text{ kg/m}^3$$

$$\text{We know } M_e = \frac{u_e}{a_e} \Rightarrow u_e = M_e \sqrt{\gamma R T_e}$$

$$= 2.3 \sqrt{1.4 \times 287 \times 145.8}$$

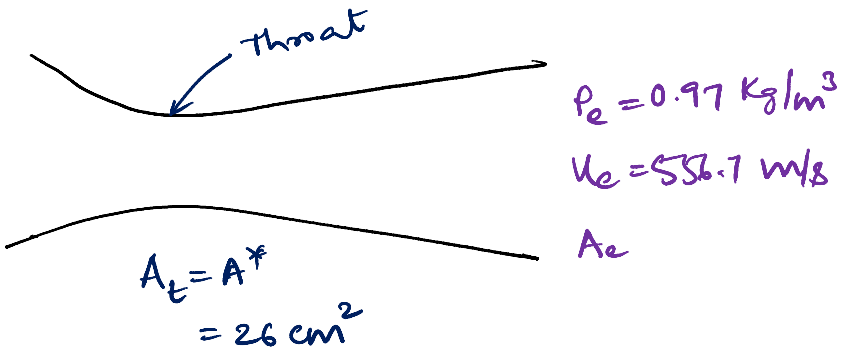
$$u_e = 556.7 \text{ m/s}$$

Problem:

In the earlier problem, the throat area is 26cm^2, find the mass flow rate through the nozzle

$\dot{m} = \rho A V$

$\dot{m} = \rho_e A_e u_e$



We know $\frac{A_e}{A^*} = 2.193$

$\Rightarrow A_e = 2.193 \times 26 \times 10^{-4}$

$= 57.02 \times 10^{-4}$ m^2

mass flow rate $\dot{m} = \rho_e A_e u_e$

$= 0.97 \times 57.02 \times 10^{-4} \times 556.7$

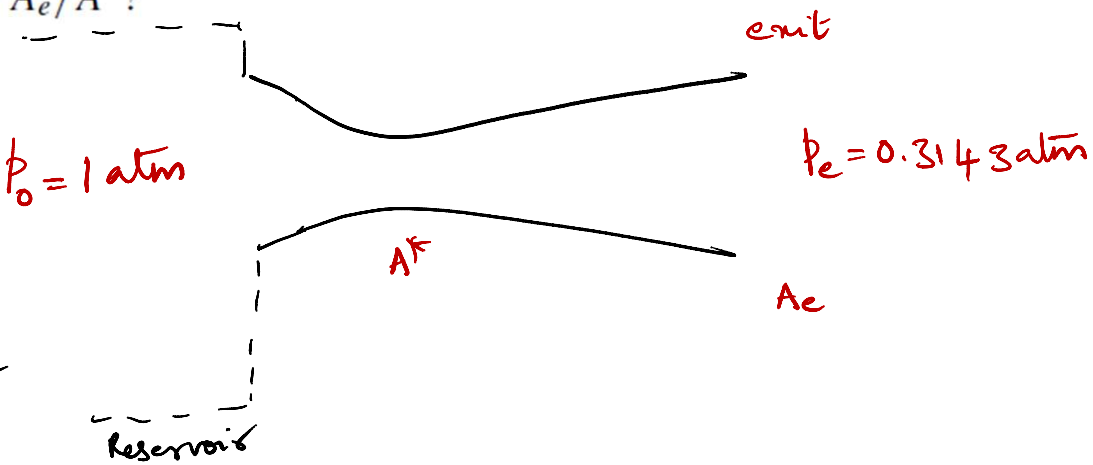
$\dot{m} = 2.92$ kg/s

$\dot{m} = \rho_t A_t u_t$

Π - method

$$\dot{m} = \frac{p_0 A^*}{\sqrt{T_0}} \sqrt{\frac{\gamma}{R} \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}}}$$

Problem: A flow is isentropically expanded to supersonic speeds in a convergent-divergent nozzle. The reservoir and exit pressures are 1 and 0.3143 atm, respectively. What is the value of A_e/A^*?



$$\frac{p_e}{p_{oe}} = \frac{p_e}{p_0} = \frac{0.3143}{1} = 0.3143$$

For $\frac{p_e}{p_{oe}} = 0.3143$ $\xrightarrow[\text{tables}]{\text{Isentropic}}$ $\frac{A_e}{A^*} = 1.115$

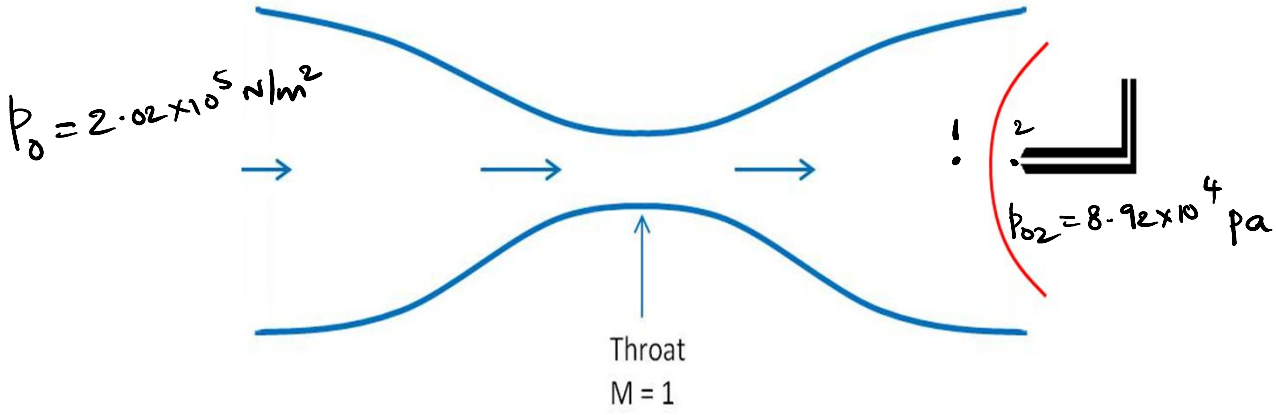
$M_e = 1.4$

Problem: A Pitot tube inserted at the exit of a supersonic nozzle reads 8.92×10^4 N/m^2. If the reservoir pressure is 2.02×10^5 N/m^2, calculate the area ratio A_e/A^* of the nozzle.

$p_{01} = p_0 = 2.02 \times 10^5$ N/m^2

$p_{02} = 8.92 \times 10^4$ N/m^2

$\frac{p_{02}}{p_{01}} = \frac{8.92 \times 10^4}{2.02 \times 10^5} = 0.441$



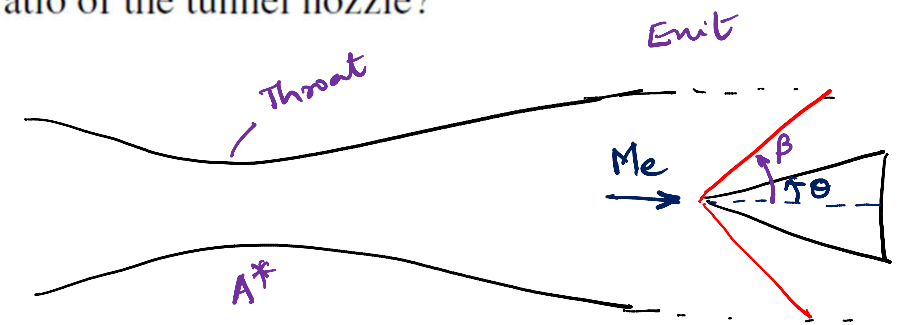
For $\frac{p_{02}}{p_{01}} = 0.441$ $\xrightarrow{\text{Normal shock tables}}$ $M_1 = 2.65$

Exit Mach number of the nozzle is $M_e = 2.65$

For $M_e = 2.65$ $\xrightarrow{\text{Isentropic tables}}$ $\frac{A_e}{A^*} = 3.306$

Problem: A 20° half-angle wedge is mounted at 0° angle of attack in the test section of a supersonic wind tunnel. When the tunnel is operating, the wave angle from the wedge leading edge is measured to be 41.8°. What is the exit-to-throat area ratio of the tunnel nozzle?

For $\frac{A_e}{A^*}$, M_e

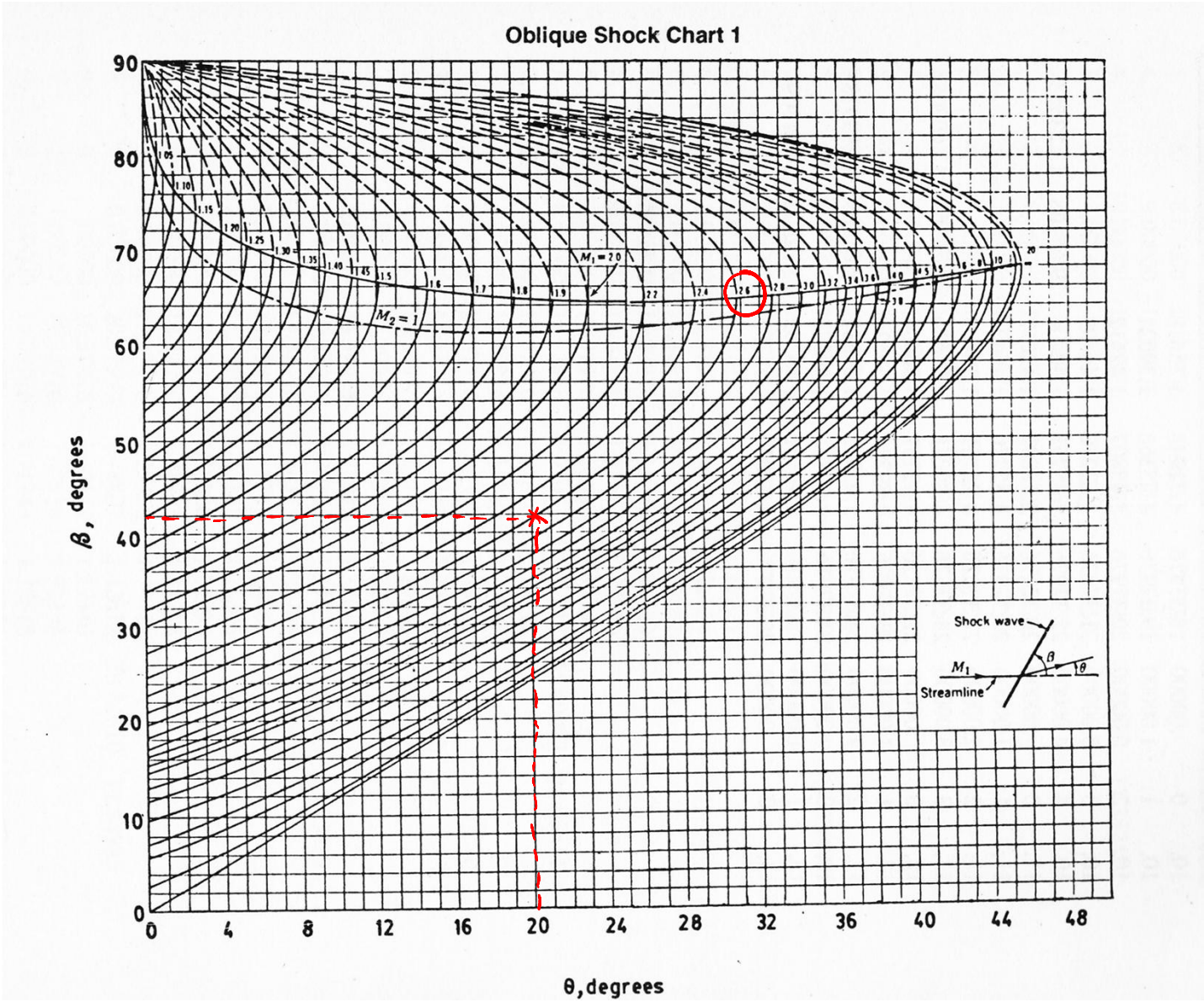


$\theta = 20^\circ$

$\beta = 41.8^\circ$

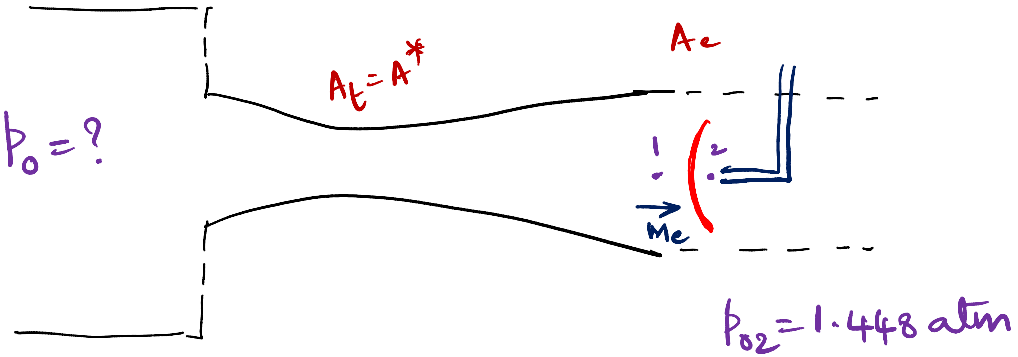
For $\theta = 20^\circ$ and $\beta = 41.8^\circ$ $\xrightarrow{\theta\text{-}\beta\text{-}M \text{ chart}}$ $M_e = 2.6$

For $M_e = 2.6$ $\xrightarrow{\text{Isentropic tables}}$ $\frac{A_e}{A^*} = 2.896$


Oblique Shock Chart 1
β, degrees
θ, degrees
0
10
20
30
40
50
60
70
80
90
0
4
8
12
16
20
24
28
32
36
40
44
48
$M_1 = 2.0$
$M_2 = 1$
Shock wave
M_1
Streamline
β
θ

Problem: The nozzle of a supersonic wind tunnel has an exit-to-throat area ratio of 6.79. When the tunnel is running, a Pitot tube mounted in the test section measures 1.448 atm. What is the reservoir pressure for the tunnel?

$$\frac{A_e}{A_t} = \frac{A_e}{A^*} = 6.79$$



Reservoir pressure $p_0 = p_{01}$

$$\frac{A_e}{A^*} \xrightarrow{\text{I.T}} M_e \xrightarrow{\text{N.S}} \frac{p_{02}}{p_{01}}$$

For $\frac{A_e}{A^*} = 6.79$ $\xrightarrow{\text{Isentropic tables}}$ $M_e = 3.5$

For $M_e = 3.5$ $\xrightarrow{\text{Normal shock tables}}$ $\frac{p_{02}}{p_{01}} = 0.213$

$$\Rightarrow p_{01} = \frac{p_{02}}{0.213} = \frac{1.448}{0.213} = 6.8 \text{ atm}$$

$$\Rightarrow \boxed{p_0 = 6.8 \text{ atm}}$$